

Skill Arena - Phase 3 Document 2

Database Schema, Financial Tables, Transaction Models & Treasury Data Architecture

Document Purpose

Define the core database structures required to support treasury management, wallets, transactions, accounting, compliance and auditing.

Architecture Philosophy

Database design must prioritize auditability, scalability, traceability and financial integrity.

Primary Database Domains

Users, Wallets, Treasury, Transactions, Tournaments, Replays, Compliance, Security and Reporting.

Users Table

Stores player identity, account status, profile references, verification status and role assignments.

Wallets Table

Stores wallet identifiers, balances, wallet types, status flags and ownership references.

Wallet Types

Live Wallet, Demo Wallet, Locked Wallet, Bonus Wallet and System Wallet.

Treasury Accounts Table

Stores reserve balances including Player Reserve, Revenue Reserve, Season Reserve, Championship Reserve, Jackpot Reserve and Emergency Reserve.

Transactions Table

Stores all financial events and references associated with deposits, withdrawals, rewards and gameplay.

Transaction Fields

Transaction ID, User ID, Amount, Currency, Type, Status, Timestamp, Source Account and Destination Account.

Ledger Entries Table

Immutable double-entry accounting records linked to treasury and wallet movements.

Ledger Fields

Ledger ID, Debit Account, Credit Account, Amount, Reference, Timestamp and Audit Identifier.

Match Financial Records

Stores entry fees, platform fees, rewards, prize allocations and treasury allocations.

Tournament Financial Records

Stores tournament fees, prize pools, allocations and championship rewards.

Reward Records

Stores XP rewards, token rewards, achievement rewards and seasonal distributions.

Audit Log Tables

Stores financial actions, approvals, administrative changes and reconciliation events.

Compliance Tables

KYC status, AML reviews, verification records, risk assessments and compliance actions.

Security Tables

Authentication events, device fingerprints, session records and incident reports.

Replay Metadata Tables

Stores replay references, replay hashes, challenge identifiers and verification records.

Reporting Tables

Aggregated metrics for finance, treasury, tournaments, users and operational reporting.

Relationships

Users own Wallets. Wallets participate in Transactions. Transactions generate Ledger Entries. Ledger Entries impact Treasury Accounts.

Scalability Requirements

Support millions of users, billions of transactions and long-term audit retention.

Data Retention Rules

Financial and audit records retained permanently unless legally required otherwise.

Backup Requirements

Automated backups, disaster recovery replication and integrity validation.

Security Requirements

Sensitive financial data encrypted at rest and protected by strict access controls.

Acceptance Criteria

Database architecture can fully reconstruct any financial state, user balance or treasury position historically.